

Quiz 9

STAT 109: Introductory Biostatistics

Quiz 9

Quiz 9 practice problems

These practice problems cover **Lectures 22–24** and **Lab 9** (tidyverse):

- **Lecture 22:** z-scores, standard normal probabilities, standardized comparisons
 - **Lecture 23:** standardized test statistic for a proportion, normal approximation, `prop.test()`, organizing analyses by variable type and purpose; **numerical** vs **categorical** variables
 - **Lecture 24:** mean and median
 - **Lab 9:** dplyr verbs and piping (`filter`, `select`, `mutate`, `summarise/summarize`, `group_by`, `count`, ...)
-

Instructions

- Work **without notes** first, then check your work.
 - Try giving these problems to your favorite AI and ask it to teach you how to do the problems and then quiz you on them. You can upload a pdf using the “+” button in Chatgpt or try Google’s Notebook LM.
-

Part A: Lecture 22 — z-scores and the standard normal

A1. Compute a z-score

Suppose women’s heights in a certain population are approximately normal with mean $\mu = 64$ inches and standard deviation $\sigma = 2.5$ inches. Annarose is 67 inches tall. Compute her **z-score** relative to this population.

A2. Compare two people with z-scores

Newton is 72 inches tall. Men’s heights in his reference population are approximately normal with mean $\mu = 70$ inches and $\sigma = 3$ inches. Compute Newton’s z-score. **Who is more unusually tall relative to their own group**, Annarose (from A1) or Newton? (Hint: be sure to compare z-scores, not raw heights.)

A3. Standard normal probability

Let Z be standard normal. Using the Standard Normal Table **table**, find $P(Z \leq 1.24)$. State what this probability represents (e.g. the **proportion of values** below $z = 1.2$ under the standard normal curve). Also find $P(Z > 1.24)$.

Part B: Lecture 23 — One proportion, z-test, organizing tools

B1. Hypotheses for a proportion

A coin is fair if the probability of heads is 0.5. You observe $x = 16$ heads in $n = 20$ flips. State H_0 and H_a for a **two-sided** test of whether the coin is fair. Identify p (population proportion of heads) in words.

B2. Standardized test statistic (concept)

In Lecture 23, the **standardized** statistic for testing a population proportion has the form

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

What role does p_0 play? (**Choose one.**)

- a) It is the observed value of $\hat{p}$ from the data.
 - b) It is the value of the population proportion or probability of a success in a single trial **assumed under** H_0 .
 - c) It is the sample size.
 - d) It is the standard deviation of the population.
-

B3. Purpose of `prop.test()` in one line

In R, `prop.test(x = 16, n = 20, p = 0.5)` tests a proportion. If you are conducting a test of a null and alternative hypothesis, what value should you look for in the output to decide whether to reject the null hypothesis in favor of the alternative? How will you decide based on the size of this value?

B4. Descriptive vs inferential

Suppose you have one **binary** variable (yes/no) from a sample. Give **one** descriptive summary you might report and **one** inferential question you might use a **confidence interval** to answer. Give an example of an inferential question you could answer with a p-value from a ****hypothesis** test**.

B5. Numerical vs categorical

For each variable below, state whether it is **numerical** or **categorical**. If it is **categorical**, also say whether it is **binary** (exactly two categories) or **not binary** (more than two categories). One line each.

- a) **Stem diameter** (cm) measured for each tree in a sample.
 - b) **Soil type** recorded as **sand**, **loam**, or **clay** for each plot.
 - c) **Parasite present** recorded as **yes** or **no** for each captured animal.
 - d) **Egg count** (0, 1, 2, ...) in each nest in a breeding survey.
-

Part C: Lecture 24 — One numerical variable (center)

C1. Mean and median by hand

Data: 3, 5, 5, 7, 9. Compute the **mean** and the **median**.

Part D: Lab 9 — tidyverse (multiple choice)

D1. Purpose of the pipe

In R, `x %>% f()` means “take `x` and pass it as the **first input** to `f()`.” What is the **main purpose** of the pipe in a tidyverse workflow? (Note: `x |> f()` has the same result)

- a) To load the tidyverse package.
 - b) To **chain** steps: send the result of one step into the next step in a natural-feeling order.
 - c) To delete missing values from a data frame.
 - d) To convert a vector to a factor.
-

D2. Subset rows by a condition

Which function **keeps only the rows** of a dataset that satisfy a logical condition such as `hwy > 30`?

- a) `select()`
 - b) `filter()`
 - c) `mutate()`
 - d) `summarise()` (aka `summarize()`)
-

D3. Select columns by name

Which function is used to **choose a subset of columns** (variables) from a dataset by name?

- a) `filter()`
 - b) `select()`
 - c) `pull()`
 - d) `arrange()`
-

D4. Collapse to summary numbers (e.g. mean)

Which function is used to **compute summary statistics** (such as one mean or one median) from a dataset, often producing **fewer rows** than the original dataset?

- a) `mutate()`
 - b) `filter()`
 - c) `summarise()` (aka `summarize()`)
 - d) `group_by()`
-

D5. New column, same length as the data

Which function **adds a new column to a dataset** in a way that the new column **contains new values but the same number of rows** as the original dataset?

- a) `summarise()`
 - b) `count()`
 - c) `mutate()`
 - d) `summarise()` with `n()`
-

D6. Count rows per category

Which function gives a **count of how many rows** fall into each category of a categorical variable?

- a) `count()`
 - b) `mutate()`
 - c) `group_by()` alone
 - d) `summarise()` without `group_by()`
-

D7. `group_by()` before summaries

What is the **role of `group_by()`** when placed **before `summarise()`**?

- a) It sorts the rows alphabetically.
 - b) It **defines groups** so that `summarise()` computes one summary number **per group**.
 - c) It drops all columns not listed.
 - d) It sorts the rows by the first column.
-

D8. `n()` inside `summarise()`

Inside `summarise()`, what does `n()` return?

- a) The number of columns in the data frame.
- b) The number of **rows** in the current group (or in the whole table if ungrouped).
- c) The maximum value of a variable.
- d) The name of the first column.

D9. Inspect a data frame

What does `glimpse(df)` (from `tibble/tidyverse`) help you see about the dataset in `df` in one quick glance? List at least two pieces of information about the dataset that are given.